

KAUSHIKAN DINAKARAN

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PROFESSIONAL SUMMARY

Data Engineer with mid-senior level experience bridging data pipelines and AI solutions. Designed an automated ingestion pipeline processing 545+ daily email reports and an LLM-powered query bot that cut weekly reporting time by 90%, while migrating legacy workflows to serverless cloud functions, reducing hosting costs 15% and execution time 30%. Skilled in building scalable ETL processes and production-ready analytics on AWS and Google Cloud. Eager to apply this expertise to deliver data-driven insights and operational efficiency for your organization.

EXPERIENCE

Think Future Technologies

Data Engineer Consultant

Jan 2024 - Nov 2025

- Architected an automated data ingestion pipeline with Python and OpenAI APIs that processed 545+ daily email status reports, extracting and centralizing text data into AWS Cloud Storage and enterprise Data Warehouses, which enabled faster access to reporting data for downstream analytics
- Engineered an LLM-powered query bot and interactive dashboards for executive leadership, enabling custom date-range queries that eliminated manual report consolidation and saved management 90% in weekly reporting time.
- Spearheaded the infrastructure migration of legacy data pipelines from AWS Lambda to Google Cloud Functions (GCF), optimizing resource allocation to cut cloud hosting costs by 15% and reduce data processing execution times by 30%
- Re-engineered and consolidated multiple client-specific Apache Airflow DAGs into dynamic, serverless AWS Lambda functions, resolving critical data bottlenecks to boost overall pipeline success rates from 80% to 95%.

Software Engineer LL

Jan 2022 - Dec 2023

- Engineered scalable backend web applications utilizing Python, the Django framework, Celery, and Redis for a US-based client, resolving critical bugs to improve system uptime and page load speeds for a platform supporting over 10 million global users.
- Developed and optimized real-time streaming data pipelines utilizing Apache Kafka, Python, and SQL, successfully processing and transforming up to 3 GB of high-throughput data per minute for international client analytics.
- Partnered with German and Israeli executive stakeholders to design and deploy business intelligence dashboards using Looker and Power BI, visualizing over 500 GB of data to enable real-time product tracking and revenue forecasting in CEO board meetings
- Led frontend visualization for an internal Robotic Process Automation (RPA) system using JavaScript, REST APIs, and D3.js, improving user interaction speed and enabling faster task completion; mentored five peer trainees on technical best practices and pipeline development

Disa Consulting Services (DCS) - "The Initiate of Your Dreams"

Internet of Things Engineer

Jul 2020 - Jun 2021

- Architected an IoT data ingestion pipeline utilizing water flow and ultrasonic sensors to capture real-time telemetry data, transmitting metrics securely to the AWS cloud to achieve near-zero latency in water usage tracking.
- Designed a centralized AWS database architecture to store and process continuous streams of sensor data, enabling automated water reservoir refilling and predictive leak detection that **eliminated manual monitoring and significantly reduced property water waste**.
- Developed a tenant-facing application with secure authentication and real-time dashboards, **empowering residents to proactively monitor unit-specific consumption and instantly identify plumbing leaks to lower monthly utility costs**.

PROJECTS

AI-Powered Automated Job Portal | AWS Lambda, GitHub Actions, S3, LLMs

- Engineered an automated data pipeline utilizing GitHub Actions and APIs to continuously extract 60–70 targeted job listings per run, securely landing the raw data into Amazon S3.
- Integrated an LLM agent to dynamically parse complex job descriptions and compare them against base profiles, generating a deterministic match score that reduced manual application and screening time by 85%.

Interactive News Research Tool | LangChain, Streamlit, Vector Retrieval [github](#)

- Architected a real-time research application utilizing LangChain and Streamlit to ingest, chunk, and vectorize up to 3 full pages of unstructured text data concurrently.
- Implemented semantic search capabilities to deliver highly accurate, contextual answers with a rapid LLM response latency of 30 to 60 seconds.

Ron - AI Emotional Support Companion | RAG, ChromaDB, Flask, Web Speech API [Github](#)

- Developed a retrieval-augmented conversational assistant utilizing semantic search over the GoEmotions dataset (58,000+ annotated Reddit comments) and 250 custom global survey responses to deliver empathic, context-aware responses.
- Deployed a lightweight Flask application featuring dual text and voice interfaces, achieving an optimized response generation time of 10 to 20 seconds during live testing with university faculty and students.

PathoVision: Medical Image Classification | ResNet, Deep Learning, Django [GitHub](#)

- Developed an automated machine learning system to classify COVID-19, pneumonia, and normal conditions by training ResNet and Inception models on a comprehensive dataset of 6,432 chest X-ray images.
- Optimized model weights and controlled overfitting through data augmentation, achieving a stable 95-96% training accuracy and ~90% validation accuracy across the 20% test split.

Respiratory Disease Statistical Analysis | Python, Statistical Modeling [Report](#)

- Analyzed massive Medical Expenditure Panel Survey (MEPS) datasets to identify asthma and influenza trends, conducting advanced Chi-Square and Mann-Whitney tests to derive insights for healthcare resource distribution.

Knowledge Management Systems Strategy | AWS QuickSight, Apache NiFi, Oracle [Report](#)

- Designed a conceptual data architecture for the manufacturing sector, integrating IoT sensors, Oracle Databases, and AWS services to facilitate enterprise-wide knowledge discovery and governance.

EDUCATION

University of North Texas

Jan 2024 - Dec 2025

Master, Data Science

Denton, TX

- **GPA:** 4/4
- **Achievements:** Awards: Recipient of the Oscar and Kaywin Blankemeyer Scholarship, Principles and Techniques for Data Science
- **Coursework:** Applied Machine Learning for Data Scientists, Data Visualization and Communication, Introduction to Health Data Analytics, Data Engineering

SRM University

2018 - 2022

Bachelor of Technology, Computer Science

- **Achievements:** Awarded a 90% Merit Scholarship for achieving top-tier scores in the highly competitive SRMJEEE.

SKILLS

- **Programming Languages:** Python, SQL, JavaScript
- **Databases & Data Warehousing:** MySQL, PostgreSQL, Snowflake, BigQuery, AWS RDS, AWS Athena, ChromaDB, SQLite
- **Data Engineering & ETL:** Apache Airflow, Talend, Apache Kafka, ETL Pipelines, Data Warehousing, Data Cleaning & Transformation
- **Cloud Platforms:** AWS (Lambda, S3, RDS, Athena, SQS), Google Cloud (GCF, BigQuery), Databricks
- **Machine Learning & AI:** TensorFlow, scikit-learn, Natural Language Processing (NLP), LLM Integration (LangChain), OpenAI APIs, Retrieval-Augmented Generation (RAG)
- **Frameworks & Libraries:** Django, Flask, FastAPI, Celery, Pandas, NumPy, Streamlit
- **Data Visualization:** Power BI, Looker Studio, Tableau, D3.js
- **Big Data & Distributed Systems:** Apache Spark, Hadoop
- **DevOps & Tools:** Docker, Git, Linux, Jupyter Notebook
- **Soft Skills:** Collaboration, Team Leadership, Problem Solving, Analytical Thinking

CERTIFICATIONS

- Learning Data Analytics
- Python for Data Science and AI
- HackerRank SQL